

COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 6

1 Artificial Intelligence (sbh11)

The standard *linear regression* model uses a hypothesis

$$h(\mathbf{x}, \mathbf{w}, b) = \mathbf{w}^T \mathbf{x} + b$$

to fit m examples $((\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m))$ by minimizing the error

$$E(\mathbf{w}, b) = \sum_{i=1}^m (y_i - h(\mathbf{x}_i, \mathbf{w}, b))^2.$$

Linear regression,
gradient descent

- (a) Derive a *gradient descent* algorithm for training the linear regression model described. [5 marks]

Answer: Start with random $\mathbf{w}$ and b and take steps $-\lambda \nabla E(\mathbf{w}, b)$ where λ denotes step size.

$$\begin{aligned} \frac{\partial E}{\partial w_j} &= \sum_i 2(y_i - h(\mathbf{x}_i, \mathbf{w}, b)) \left(-\frac{\partial(\mathbf{w}^T \mathbf{x}_i + b)}{\partial w_j} \right) \\ &= -\sum_i 2(y_i - h(\mathbf{x}_i, \mathbf{w}, b)) \mathbf{x}_i^{(j)} \end{aligned}$$

and

$$\begin{aligned} \frac{\partial E}{\partial b} &= \sum_i 2(y_i - h(\mathbf{x}_i, \mathbf{w}, b)) \left(-\frac{\partial(\mathbf{w}^T \mathbf{x}_i + b)}{\partial b} \right) \\ &= -\sum_i 2(y_i - h(\mathbf{x}_i, \mathbf{w}, b)) \end{aligned}$$

Linear regression,
gradient descent

- (b) In the application of interest, you believe that it is desirable to train such that the learned parameters have $\|\mathbf{w}\| \simeq 1$. Suggest a modification to $E(\mathbf{w}, b)$ that facilitates this, and derive the corresponding gradient descent training algorithm. [5 marks]

Answer: Not specifically covered, but requiring only minor creativity, we use

$$E(\mathbf{w}, b) = \sum_{i=1}^m (y_i - h(\mathbf{x}_i, \mathbf{w}, b))^2 + \theta(\|\mathbf{w}\|^2 - 1)^2$$

where θ sets how much we want to enforce the norm of $\|\mathbf{w}\|$. The derivation then leaves the gradient for b unchanged, and adds a term to the gradient for w_j

$$\begin{aligned} \frac{\partial}{\partial w_j} \theta(\|\mathbf{w}\|^2 - 1)^2 &= 2\theta(\|\mathbf{w}\|^2 - 1) \frac{\partial}{\partial w_j} (\|\mathbf{w}\|^2 - 1) \\ &= 2\theta(\|\mathbf{w}\|^2 - 1) \frac{\partial}{\partial w_j} \left(\sum_k w_k^2 - 1 \right) \\ &= 4\theta(\|\mathbf{w}\|^2 - 1) w_j. \end{aligned}$$

Heuristic search (c) Describe the components of a general *heuristic search* problem. [4 marks]

Answer: We have states, which can be expanded into one or more child states. We have a start state and one or more goal states. There is a cost for moving from any state to a successor. There is a heuristic that estimates the distance from any state to an optimal goal.

Heuristic search,
machine learning (d) You are faced with a heuristic search problem, but the heuristics you have so far developed are less effective than desired. Suggest two ways in which supervised machine learning might be used to develop a better heuristic, mentioning if necessary any corresponding disadvantages of using the approach. You may assume that a collection of problems to be solved by the heuristic search is available. [6 marks]

Answer: This is open-ended, and any sensible suggestions will be accepted. For example:

- By running the search on the collection of problems you obtain a collection of search graphs that can be analyzed to learn the actual path costs from some states to goals. If the states can be described with features then this provides a training set allowing a learner be used directly as a heuristic.
- Alternatively, the existing heuristics can be used as features. A similar data gathering exercise allows the learner, again, to be used to combine existing heuristics into a, hopefully, better alternative.

The reason for asking for disadvantages is that some appreciation should be shown for the fact that heuristics ought to be monotonic, and this may be hard to guarantee in both scenarios.
